



Learning Constitutive Laws in 2D Granular Flow via Neural Network Frameworks

ME 299: Undergraduate Research Project Report

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1 Introduction and Objectives

The primary objective of this research is to establish a robust neural network framework capable of autonomously learning the constitutive laws governing 2D granular flows. Unlike Newtonian fluids, granular materials exhibit complex rheological behaviors—including non-local effects and pressure-dependent friction—that make traditional analytical modeling computationally expensive and theoretically limited.

This study follows a bottom-up approach: first validating the neural network architecture on fundamental mathematical functions, then transitioning to noisy 2D simulation data from the Discrete Element Method (DEM). The ultimate goal is to create a computational model that can replace time-intensive simulations with real-time stress field predictions.

2 Phase I: Fundamental Function Approximation

Before engaging with complex granular physics, the framework was benchmarked on non-linear mathematical functions to determine the minimum complexity required for the network architecture.

2.1 Architectural Selection: The 3-Layer Rule

We systematically tested networks ranging from 1 to 5 hidden layers. It was observed that 1-2 layer "shallow" networks significantly underfit the data, failing to capture the curvature of quadratic functions and the periodicity of sine waves. Conversely, architectures with more than 4 layers provided diminishing returns in accuracy while increasing the risk of overfitting.

Key Finding: A 3-layer network (20 neurons per hidden layer) provided the optimal balance, achieving a low prediction error on quadratic and sine approximations.

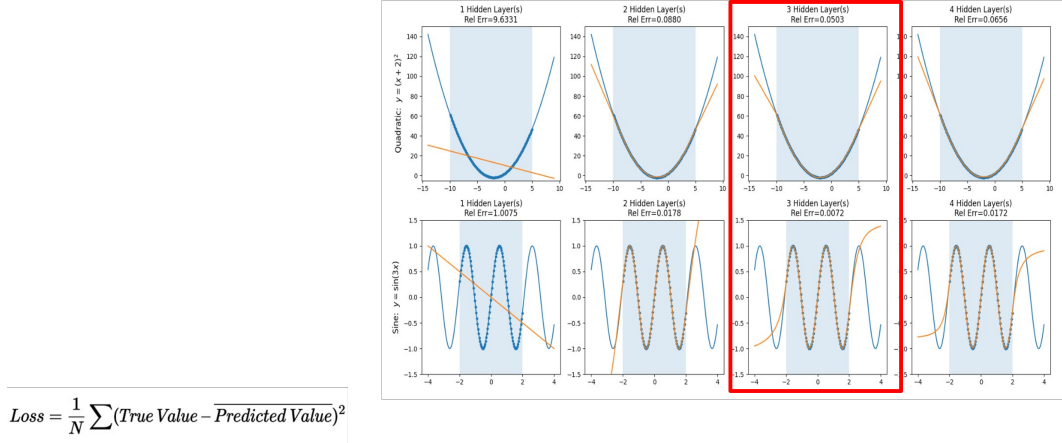


Figure 1: Evolution of function fit as a function of network depth for Quadratic and Sine functions.

3 Phase II: Architectural Tournament and Optimization

To identify the optimal model, we conducted a systematic tournament involving 25 distinct hyperparameter combinations. This included 5 activation functions (ReLU, Tanh, Swish, Mish, GELU) and 5 loss functions (MSE, MAE, Huber, MSLE, RE).

3.1 Activation and Loss Benchmarking

- **Activation Functions:** While ReLU is excellent for piecewise linear behavior, it struggled with the oscillatory nature of stress fluctuations. Tanh and Mish emerged as superior candidates due to their smooth, continuous gradients.
- **Loss Functions:** Relative Error and Huber Loss were more robust than Mean Squared Error when handling granular data, which often contains outliers due to localized particle collisions.

4 Phase III: Granular Data Processing Pipeline

The transition to real physics involved processing data from LAMMPS simulations of 2D granular flow. This data is naturally noisy due to the discrete nature of particles.

4.1 Quasi-Linear Masking

To extract meaningful constitutive laws, we implemented a rolling standard deviation mask. This algorithm scans the vertical stress profiles and selects only the "quasi-linear" regions—zones where the flow is stable and the stress-gradient is consistent. This effectively filtered out wall-boundary interference and transient dead zones near the fixed boundary.

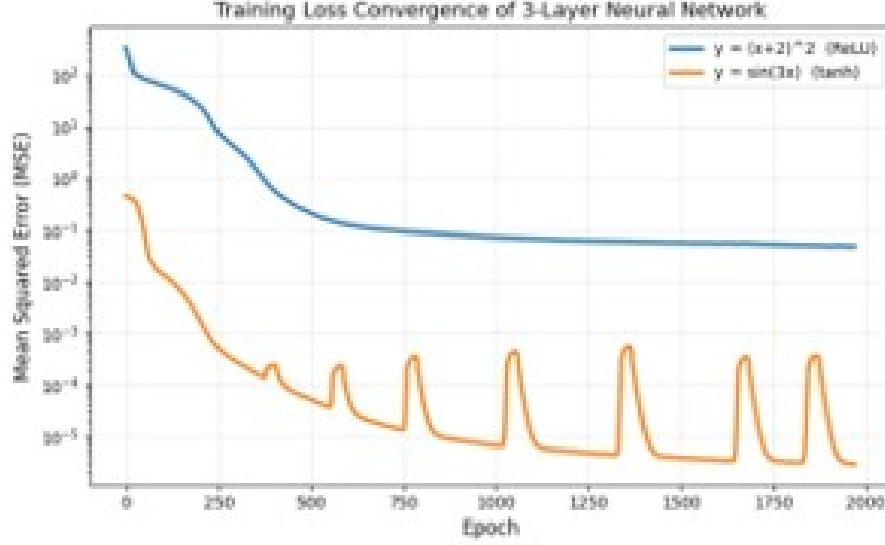


Figure 2: Training loss convergence behavior using AdamW and Cosine Annealing optimization.

4.2 Reference Normalization Strategy

Initial deployment across different datasets showed high errors because the model was sensitive to absolute stress magnitudes. We developed a breakthrough reference normalization method:

$$\sigma_{norm} = \frac{\sigma_{raw}}{\bar{\sigma}_{dataset}} \quad ; \quad \dot{\gamma}_{norm} = \frac{\dot{\gamma}_{raw}}{\dot{\gamma}_{dataset}} \quad (1)$$

By scaling inputs and outputs by the dataset means, we successfully decoupled the learned "material physics" from the "environment scale."

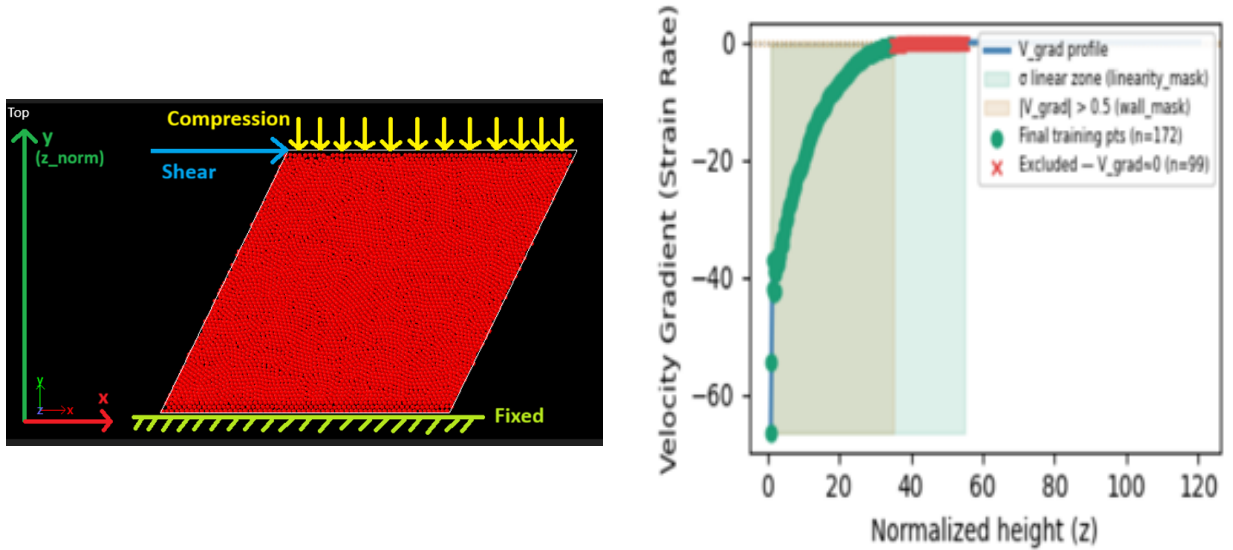


Figure 3: Left: LAMMPS simulation setup. Right: Velocity gradient masking for stable training zone selection.

5 Phase IV: Performance Analysis and Validation

Validation was performed on two distinct levels: cross-dataset interpolation (Validation Set 1) and environmental regime shift (Validation Set 2).

5.1 Validation Set 1: Scaling Success

By applying the reference normalization learned during training to a completely new dataset, the model maintained high fidelity. The prediction error dropped from 48% (unscaled) to approximately 3%. This proves that the neural network successfully learned the ratio of stress to strain rate (the constitutive law) regardless of absolute pressure values.

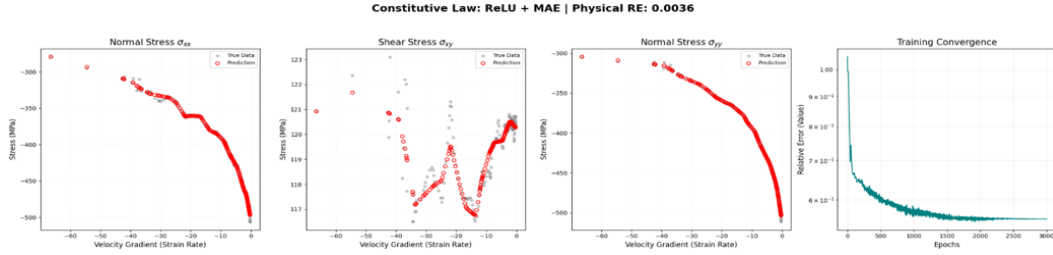


Figure 4: Constitutive Validation: Model Prediction vs. DEM Ground Truth after normalization.

5.2 Validation Set 2: The Gravity-Shift Challenge

The model was tested on a zero-gravity bulk simulation. In this regime, the source of pressure changes from gravitational weight to purely boundary-driven compression.

- **Normal Stresses** (σ_{xx}, σ_{yy}): Remained accurate with errors under 14%, as they directly reflect local inter-particle confinement.
- **Shear Stress** (σ_{xy}): Error diverged significantly to approximately 84.56%.

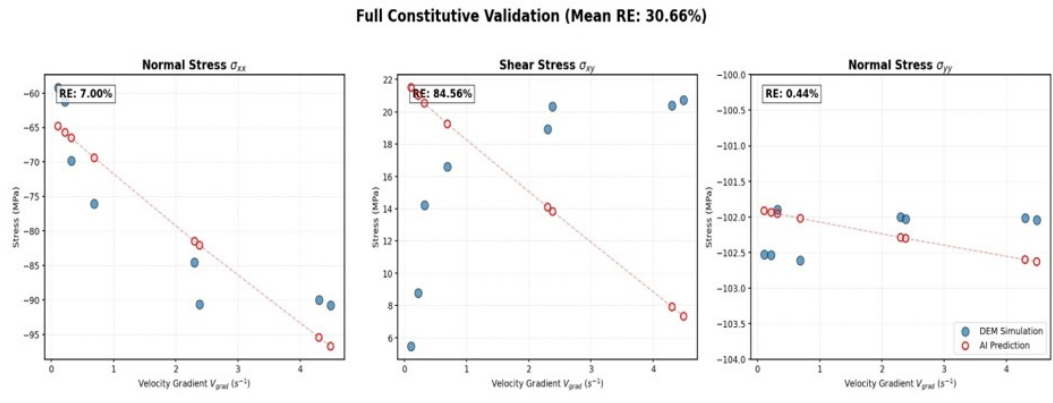


Figure 5: Constitutive Validation: System performance in the absence of gravity highlighting shear stress divergence.

6 Physical Interpretation: Pressure-Source Decoupling

The divergence in shear stress magnitude during the zero-gravity test provides a profound physical insight termed pressure-source decoupling.

The neural network was trained on a system where pressure increases with depth due to gravity. Consequently, the model learned a hidden correlation between depth and shear resistance. When gravity is removed, this correlation breaks. While the model correctly predicts the qualitative trend, it fails to predict the absolute magnitude because pressure was not explicitly provided as an input feature. This confirms that granular constitutive laws are intrinsically linked to the confining pressure environment.

7 Conclusion and Future Directives

The developed framework successfully learned the underlying constitutive patterns of 2D granular flows. We demonstrated that with proper statistical masking and reference normalization, deep learning models can generalize across different simulation scales with high accuracy.

7.1 Key Achievements

- a) **Optimal Architecture:** Identified the 3-layer (20 neurons per hidden layer) Tanh/Mish configuration as the optimal choice for mechanical system modeling.
- b) **Normalization:** Reduced cross-dataset error from 48% to 3% using mean-reference scaling.
- c) **Physical Insight:** Quantified the impact of the pressure-source (gravity vs. boundary) on constitutive law learning.

7.2 Future Work

- **Physics-Informed Neural Networks:** Incorporate conservation of momentum directly into the loss function to bound shear stress predictions.
- **Variable Pressure Inputs:** Include the local pressure as an explicit input to the network to resolve the pressure-source decoupling issue found in zero-gravity tests.